



DEEP SLEEP

French painter Henri Rousseau (1844-1910) introduced unreal, dreamlike, and often slightly comic qualities to much of his work, like *The Sleeping Gypsy*. A mixture of recognizable figures in strange situations and odd settings, often from the very recent past, occurs in the dreams of many people. One explanation is that the dreaming brain is fast-replaying recent events. It stores "significant" ones in the memory banks and discards others, even if the significance is not obvious to the dreamer. Dreams seem to happen at a deep, primitive level and have been called the "core of the unconscious."

DEEP THOUGHT

Auguste Rodin (p. 6) portrayed pensive concentration in his statue of *The Thinker*. Recent research shows that conscious awareness of our surroundings is largely based on sight (pp. 52-53). When humans want to concentrate on thought processes, they "stare into space," almost unseeing, so that inner mental processes come to the fore. This phenomenon is almost unique to humans.



MATTERS OF THE MIND

One field of medicine to emerge recently is psychiatry, pioneered by Austrian Sigmund Freud (1856-1939). It involves the study and treatment of mental and emotional problems, especially those deep in the unconscious, apart from everyday thinking and awareness. Its counterpart study of the mind's "normal" processes, and the behavior that results, is called psychology. Great progress has been made in linking mental disorders to abnormalities of structure or chemistry in the brain. Some day thoughts might be measured as a Descartes-like physical system of electrical signals traveling along nerves.

ALTERNATIVE APPROACH

It is possible that there could be a nonphysical area of the human mind that cannot be measured by today's instruments or described as nerve signals and chemicals. The mind may be able to transcend matter, with techniques such as meditation. With so much anatomical and physiological knowledge, these are questions for future body scientists.

